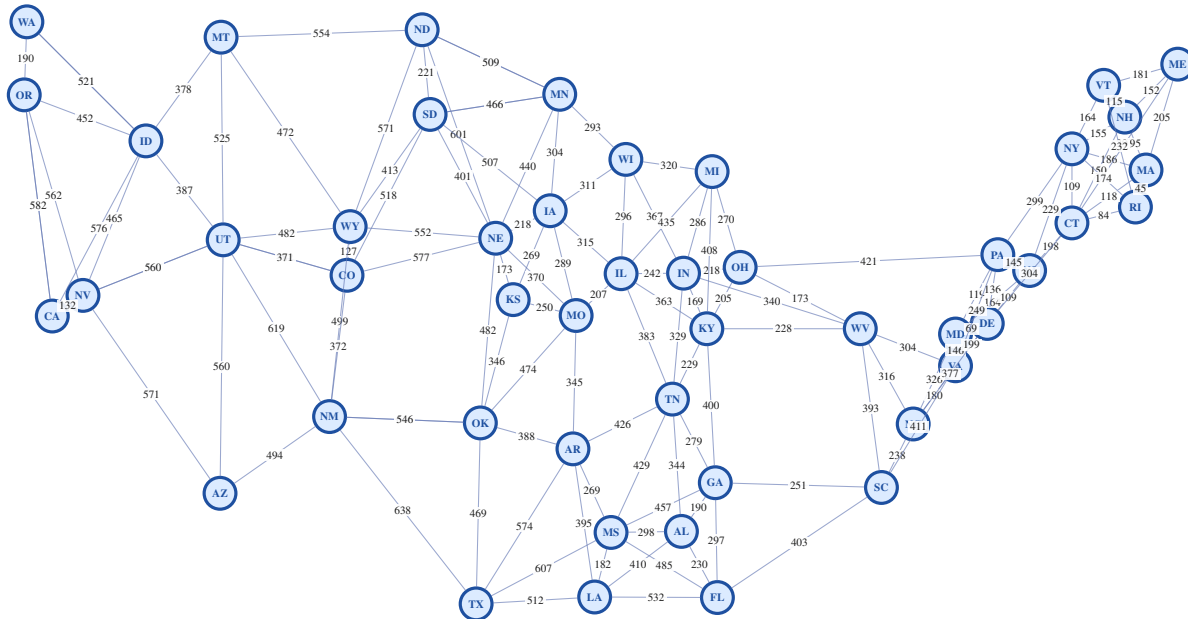


Goals:

- Understand the terms: tree, spanning tree, minimum cost spanning tree
- Understand how to use Kruskal's Algorithm to find a minimum cost spanning tree
- Know of applications of minimum cost spanning trees

Weighted Road Graph: 48 Contiguous U.S. State Capitals

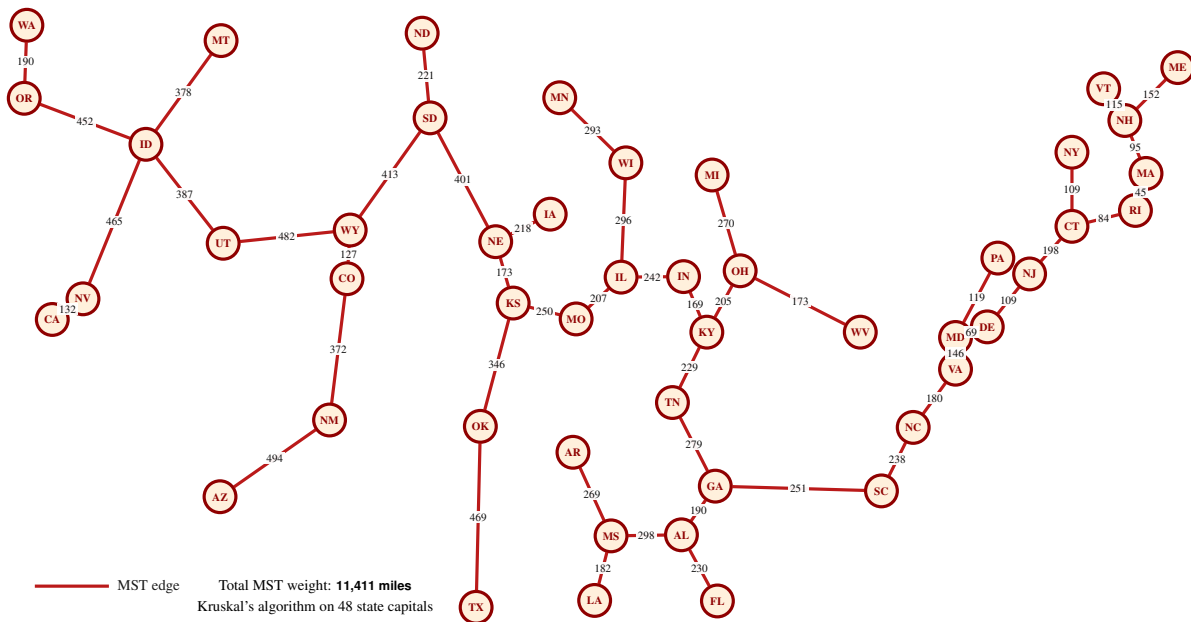
Edge weights = approximate road distance in miles. Each capital connected to its 4–5 nearest neighbors.



What do we notice?

Minimum Cost Spanning Tree (Kruskal's Algorithm)

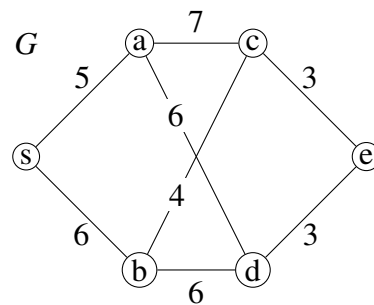
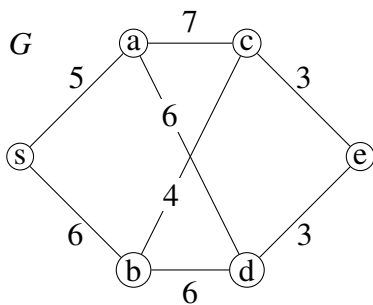
Total MST weight $\approx 11,411$ miles. Red edges form the unique spanning tree of minimum total road distance.



1. Definitions

(a) **weighted graph**(b) **tree**(c) **spanning tree**(d) **minimum cost spanning tree** (minimum weight spanning tree)

2. Example:



3. Kruskal's Algorithm

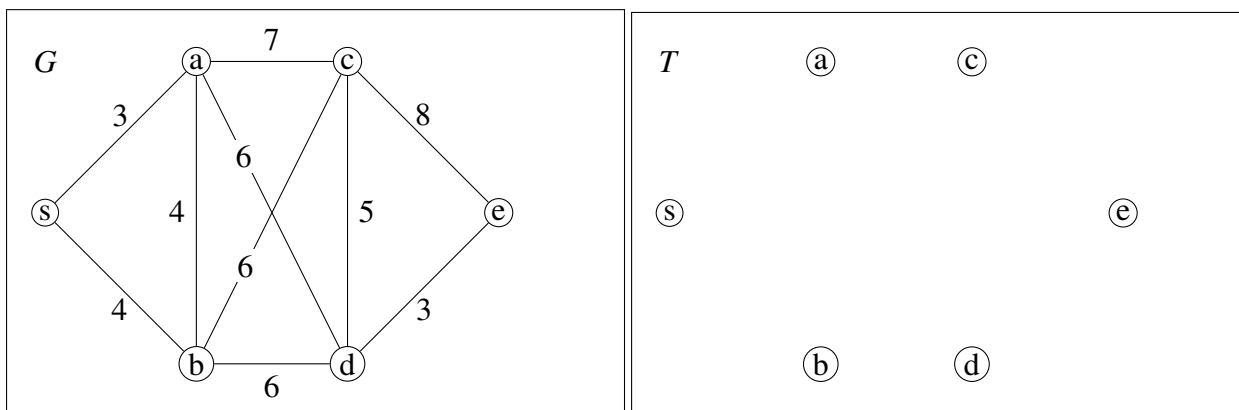
input: a graph, G , with costs (or weights) on the edges

output: a spanning tree, T , of minimum cost

Steps:

- (a) (Initialization Step:) T is a graph on the vertex set of G but with no edges.
- (b) (Iterative Step:)
 - i. Select the cheapest unused edge in the graph. (Ties are broken alphabetically.)
 - ii. If the edge does **not** create a cycle, add the edge to T . Otherwise, reject the edge.
 - iii. Mark the edge as used.
 - iv. If T is a spanning tree, STOP. Otherwise return to the beginning of the iterative step.

4. Use Kruskal's Algorithm to find the minimum cost spanning tree for the graph G below.



Used?	edges	weights

5. Think of an application of Kruskal's Algorithm.